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SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Cutting Edge, check out the future of wireless technology. Then, in Space Age, an update on the James Webb Telescope. Finally, in R&D, we look at a new test that can detect over 50 genetic diseases.

Source: MIT News

Clothes have ears

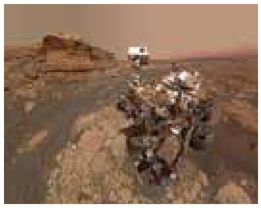
A new fabric with the ability to record sounds has been developed by a team of researchers. This may seem to be something straight out of the comic books, but the researchers, in their paper titled, "Single fiber enables acoustic fabrics via nanometer-scale vibrations", elucidated all the possibilities that this fiber could unlock. **https://dgit.in/fabricears**

WHAT'S NEW

Sounds of Mars

Recent research published in the journal Nature, carried out by a multinational team of scientists, has brought to light several facts related to the ambient noise of our red neighbour, Mars. The SuperCam instrument installed on the Perseverance rover, which has been exploring the planet's surface, had recorded these sounds on the 19th of February in 2021, which was just a day after it had landed on Mars.

Previously, we had received various images from the planet, giving us an idea of the terrain and physical features of Mars. Still, the question that had remained was about the ambient sound of the red planet. Well, as it turns out, the planet is relatively very silent.

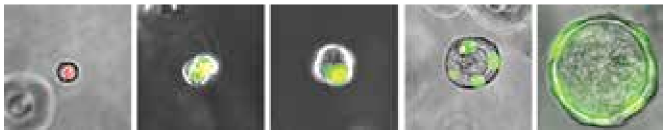


Further exploration of the hours of audio captured, as published in the paper titled, "In situ recording of Mars soundscape", revealed that the planet is primarily silent, with natural sound sources apart from the wind being relatively absent.

The paper also revealed that the speed of sound on Mars is slower than that on planet Earth. The speed was measured to be about 240 m/s on Mars, which is 100 m/s slower than the recorded speed of sound on Earth, which is 340 m/s. **https://dgit.in/marsound**

New cells in our bodies

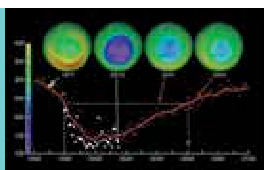
A team of researchers from the Perelman School of Medicine at the University of Pennsylvania have found a new cell in the human lungs.



Source: Penn Medicine

The team's research findings were published on the 30th of March this year in Nature, titled, "Human distal airways contain a multipotent secretory cell that can regenerate alveoli". These new cells, called respiratory airway secretory cells (RASCs), were identified during the team's study of the human lung tissues.

These newly discovered cells were located very deep inside the lungs, making up the lining of the tiny air way branches, which exist near the alveoli structures. These structures are the site of gaseous exchange in the lungs, which means that further probing into these cells would prove vital in studying various lung-related diseases in humans. These cells have been found to have stem cell-like properties, which help them regenerate. **https://dgit.in/newlungcell**



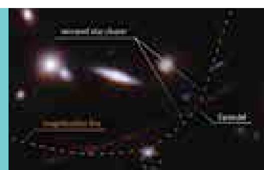
Hotter than expected

Ozone gas was found to be a much more significant contributor to heating up the Earth's atmosphere than initially thought. The study revealed that the changes in the levels of Ozone were the reason behind the rise in temperature around Antarctica. **https://dgit.in/hotozone**

Source: Nature

Upbringing = Navigational capability

Research carried out by a team of neuroscientists, with about 400,000 people from 38 countries studying their navigational capabilities has shown that the navigational capabilities of humans are dependent upon the region they grew up in. **https://dgit.in/humnav**



'Earendel': The farthest star

We may have just looked at the farthest star from us to date. It is so far that the light from that star took about 12.9 billion years in total to reach us. The previous record was ~8.9 billion years. Named 'Earendel', it is set to be explored further using the JWT. **https://dgit.in/earandel**